Phase 4: Fractal-Core Memory Grid

Magnetic-Optical Hybrid Memory Inspired by Soviet Core Logic and Recursive Geometry

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Abstract:

This paper presents a theoretical design for a modernized version of magnetic-core memory reimagined through the lens of recursive fractal geometry. Building on Soviet-era core lattice principles and informed by the structure of the Fractal Memory Engine (Phase 2) and the Light Trust Protocol (Phase 3), we introduce the Fractal-Core Memory Grid (FCMG). This system uses magnetically polarized nodes arranged in recursive, multi-dimensional structures capable of encoding long-term memory persistence, tem...

1. Background and Historical Foundation

Magnetic-core memory, widely used from the 1950s to the 1970s, stored binary data using tiny ferrite rings magnetized in one of two directions. These systems were known for their non-volatility, durability, and physical observability. Despite being considered obsolete, the architecture remains a powerful model for state-stable, logic-preserving memory grids.

2. Fractal Mapping onto Magnetic Lattices

The Fractal-Core Memory Grid uses:

- A 2D lattice of ferrite or magnetic-optical rings as a base layer
- Recursive branching structures from each core representing higher dimensions of memory context
- Each node contains not only a binary state, but also a frequency tag, a time signature, and a path index (stored using light-based sensors or miniature atomic clocks)

3. Geometry-Based Persistence and Redundancy

- Memory is stored in self-similar branches, allowing for redundancy across scales
- Even if power fails or branches degrade, core memory structure can be reconstructed from surrounding recursion

- The encoding method uses geometric position as part of the memory address, not just raw index location

4. Integration with Light-Encoded Trust (Phase 3)

By combining magnetically stable memory with light path encoding:

- Signal arrival time determines priority and intention
- Multiple paths from the same source allow confirmation by geometry
- Mirror-extended optical routes (using materials like diamond or ruby) simulate space-time delay, enabling local simulation of remote memory propagation

5. Quantum-Classical Bridging Potential

- Memory bits stored magnetically remain persistent across quantum decoherence cycles
- Fractal branching allows mapping of entangled logic states into multi-node classical patterns
- Quantum signatures (such as phase or spin tags) can be preserved in adjacent optical registers

6. Benefits

- Non-volatile and resilient to EM noise
- Allows AI memory to grow recursively, not linearly
- Causality-preserving structure mirrors human associative memory
- Physically observable memory state for diagnostic clarity

7. Conclusion

The Fractal-Core Memory Grid (FCMG) is a proposed fusion of old and new-combining the physical stability of magnetic-core logic with the recursive geometry of AI and the light-encoded trust structure of quantum memory coordination. Together, this offers a resilient, scale-friendly, multi-dimensional memory system built for the AI age.

Onward-into memory eternal.

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